

Titanic Dataset — Project 1 Documentation

Data Cleaning & Preparation

DecodeLabs Industrial Training Kit — Data Analytics Project 1

Submitted By

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Domain: Data Analytics

Project: Data Cleaning & Preparation

Objective

The objective of this project is to clean and prepare the Titanic dataset for further analysis and Machine Learning applications. The dataset contained missing values, formatting inconsistencies, and unnecessary columns. Data cleaning was performed to improve data quality and ensure reliable analysis.

Dataset Information

Dataset Name

Titanic Dataset

Source

Kaggle Titanic Dataset

Dataset Description

The Titanic dataset contains passenger information such as:

- Passenger ID
- Name
- Age
- Gender
- Ticket Fare
- Passenger Class

- Survival Status
- Embarkation Port

This dataset is widely used for data analysis and Machine Learning practice.

Tools & Technologies Used

Tool	Purpose
Python	Data Cleaning
Pandas	Data Manipulation
NumPy	Numerical Operations
Jupyter Notebook	Project Development

Importing Required Libraries

```
import pandas as pd  
import numpy as np
```

Loading the Dataset

```
df = pd.read_csv("Titanic-Dataset.csv")
```

Initial Dataset Inspection

Viewing First Five Rows

```
df.head()
```

Checking Dataset Shape

```
df.shape
```

Checking Column Names

```
df.columns
```

Dataset Information

```
df.info()
```

Problems Identified in the Dataset

After inspecting the dataset, the following issues were identified:

1. Missing values in Age column
2. Missing values in Embarked column
3. Large number of missing values in Cabin column
4. Possible duplicate rows
5. Inconsistent text formatting

Missing Value Handling

Checking Missing Values

```
df.isnull().sum()
```

Handling Missing Values in Age Column

The Age column is numerical, so missing values were filled using the median.

```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

Why Median?

Median is less affected by outliers compared to mean and provides better results for skewed distributions.

Handling Missing Values in Embarked Column

The Embarked column is categorical, so missing values were filled using the mode.

```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

Why Mode?

Mode represents the most frequent category and is commonly used for categorical missing values.

Removing Cabin Column

The Cabin column contained too many missing values, so it was removed from the dataset.

```
df = df.drop("Cabin", axis=1)
```

Verifying Missing Values

```
df.isnull().sum()
```

Result

All important columns were successfully cleaned and missing values were handled.

Duplicate Row Detection

Checking Duplicate Rows

```
df.duplicated().sum()
```

Removing Duplicate Rows

```
df = df.drop_duplicates()
```

Result

Duplicate rows were removed successfully.

Text Formatting & Cleaning

Removing Extra Spaces from Name Column

```
df["Name"] = df["Name"].str.strip()
```

Converting Gender Values to Lowercase

```
df["Sex"] = df["Sex"].str.lower()
```

Purpose

Standardized text formatting improves consistency and prevents analysis errors.

Data Type Verification

Checking Data Types

```
df.dtypes
```

Observation

The dataset contained correct data types for numerical and categorical columns.

Final Dataset Verification

Checking Missing Values Again

```
df.isnull().sum()
```

Checking Duplicate Rows Again

```
df.duplicated().sum()
```

Final Result

- No missing values in important columns
 - No duplicate rows
 - Proper formatting achieved
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Saving the Cleaned Dataset

```
df.to_csv("cleaned_titanic.csv", index=False)
```

Before Cleaning vs After Cleaning

Issue	Before Cleaning	After Cleaning
Missing Values	Present	Removed/Handled
Duplicate Rows	Possible	Removed
Text Formatting	Inconsistent	Standardized
Dataset Quality	Raw	Cleaned

Key Learnings

Through this project, the following skills were learned:

- Data Cleaning
- Missing Value Handling

- Duplicate Removal
 - Text Formatting
 - Data Preparation
 - Pandas Operations
 - Professional Data Workflow
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Conclusion

The Titanic dataset was successfully cleaned and prepared for analysis. Missing values were handled using statistical techniques such as median and mode. Duplicate rows were removed, and text formatting issues were corrected.

The cleaned dataset is now reliable and ready for Exploratory Data Analysis (EDA) and Machine Learning applications.

This project provided practical experience in real-world data preprocessing and improved understanding of data quality management.

Future Scope

The cleaned dataset can now be used for:

- Exploratory Data Analysis (EDA)
 - Data Visualization
 - Machine Learning Models
 - Survival Prediction
 - Statistical Analysis
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Thank You

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